PROJECT PLAN

QueryMind AI

A Natural Language Database Assistant

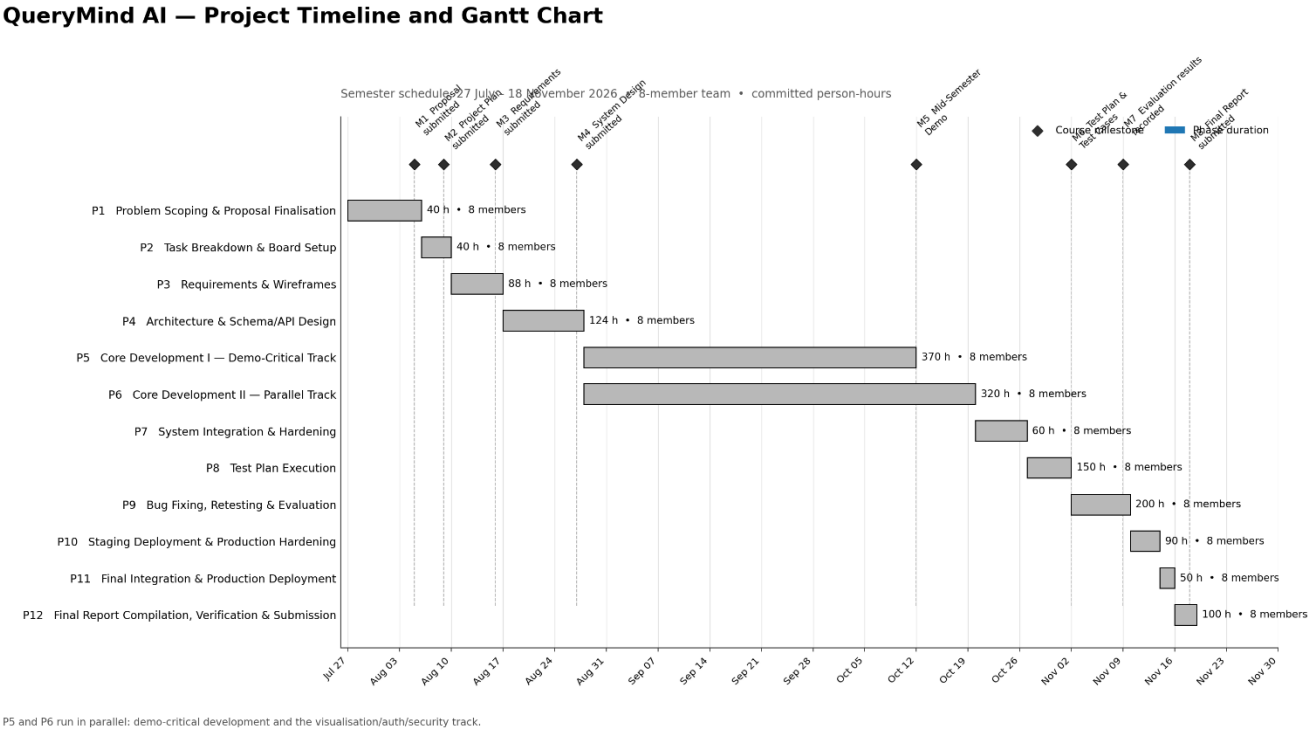
1. Team Roster

Member	Student ID	Responsibility Area
Deep Soni	202512089	Deployment and Post Integration Testing
Ishpreet Kaur	202512104	Backend / Database and Query Execution
Neel Shah	202512114	Backend / Database and Query Execution
Harsh Rathod	202512086	Security and Query Validation
Nishit Gal	202512002	Security and Query Validation
Parshwa Jain	202512096	Backend / Database and Query Execution
Heer Sachdev	202512117	Frontend / UI and UX
Bansari Akhani	202512032	Frontend / UI and UX

2. Complete Team Timeline

Phase	Name	Start	End	Duration	Hours
P1	Problem Scoping & Proposal Finalisation	2026-07-27	2026-08-05	10d	40
P2	Task Breakdown & Board Setup	2026-08-06	2026-08-09	4d	40
P3	Requirements & Wireframes	2026-08-10	2026-08-16	7d	88
P4	Architecture & Schema/API Design	2026-08-17	2026-08-27	11d	124
P5	Core Development I - Demo-Critical Track	2026-08-28	2026-10-11	45d	370
P6	Core Development II - Parallel Track	2026-08-28	2026-10-19	53d	320
P7	System Integration & Hardening	2026-10-20	2026-10-26	7d	60
P8	Test Plan Execution	2026-10-27	2026-11-01	6d	150
P9	Bug Fixing, Retesting & Evaluation	2026-11-02	2026-11-09	8d	200
P10	Staging Deployment & Production Hardening	2026-11-10	2026-11-13	4d	90
P11	Final Integration & Production Deployment	2026-11-14	2026-11-15	2d	50
P12	Final Report Compilation, Verification & Submission	2026-11-16	2026-11-18	3d	100

2.1 Team Gantt Chart



P5 and P6 run in parallel: demo-critical development and the visualisation/auth/security track.

2.2 Milestones on the Timeline

Official submission deadlines:

- 5 Aug 2026 - Project Proposal
- 9 Aug 2026 - Project Plan
- 16 Aug 2026 - Requirements Specification
- 27 Aug 2026 - System Design
- Week of 12 Oct 2026 - Mid-Semester Demonstration
- Week of 2 Nov 2026 - Test Plan and Test Cases
- Week of 16 Nov 2026 - Final Application
- Week of 16 Nov 2026 - Deployment & Hosting
- 18 Nov 2026 - Final Project Report
- 18 Nov 2026 onwards - Final Presentation & Demo
- 19 Nov 2026 - Peer Review & Contribution Report
- Weekly, 27 Jul - 15 Nov 2026 - Progress Reports

Internal completion milestones:

- 11 Oct 2026 - Demo-critical pipeline (query understanding, execution, chat UI) feature-complete, ahead of the demo week
- 19 Oct 2026 - Parallel track (visualisation, auth/RBAC, admin panel, security policy) complete
- 26 Oct 2026 - Full system integration & hardening complete; staging-ready build frozen
- 1 Nov 2026 - Formal testing (unit/integration/security/UI/DB) complete; test documentation finalized - before the Test Plan submission week begins
- 9 Nov 2026 - Bug fixing, retesting, and evaluation-set accuracy results recorded
- 13 Nov 2026 - Staging deployment verified; configuration/environment issues resolved
- 15 Nov 2026 - Production deployment live, final integration signed off - ahead of the Week-of-16-Nov deadline
- 18 Nov 2026 - Final report compiled, verified, submitted

3. Phase Schedule and Milestones

Phase	Dates	Main Work	Owner(s)	Hours	Milestone / Deliverable
P1	27 Jul-5 Aug	Proposal scoping (all areas)	All members	40	Proposal submitted (5 Aug)
P2	6-9 Aug	Backlog & board setup	All members	40	Project Plan submitted (9 Aug); Trello live
P3	10-16 Aug	Requirements, user stories, wireframes	All members	88	Requirements Spec submitted (16 Aug)
P4	17-27 Aug	Architecture, MongoDB schema, API contract	All members	124	System Design submitted (27 Aug); schema/API frozen
P5	28 Aug-11 Oct	NLP/text-to-query, schema discovery, query execution, chat UI	Parshwa Jain, Ishpreet Kaur, Neel Shah, Heer Sachdev	370	Demo-critical pipeline complete (11 Oct)
P6	28 Aug-19 Oct	Visualisation/export, auth/RBAC, admin panel, security policy (parallel to P5)	Bansari Akhani, Nishit Gal, Harsh Rathod	320	Parallel track complete (19 Oct)
P7	20-26 Oct	Merge all modules; feature freeze; staging environment scaffolding	Deep Soni (lead), all members	60	Integration & hardening complete (26 Oct)
P8	27 Oct-1 Nov	Unit, integration, security, UI, DB testing; test docs compiled	Deep Soni (lead), Harsh Rathod	150	Formal testing complete; Test Plan submitted (wk of 2 Nov)
P9	2-9 Nov	Bug fixing, retesting, 150-pair accuracy evaluation	Deep Soni (lead), all members	200	Retesting & evaluation complete (9 Nov)
P10	10-13 Nov	Staging deployment, AWS config, environment hardening, docs	Deep Soni (lead)	90	Staging verified (13 Nov)
P11	14-15 Nov	Production deployment; final integration sign-off	Deep Soni (lead)	50	Production live (15 Nov); ahead of wk-of-16-Nov deadline
P12	16-18 Nov	Report compilation, verification, formatting	Deep Soni (lead), all members	100	Final Report/Presentation submitted (18 Nov)

4. Team Task Assignments

Member	Student ID	Responsibility Area	Major Tasks	Dependencies	Major Milestone
Deep Soni	202512089	Deployment and Post Integration Testing	Evaluation set, integration coordination, test-plan execution lead, staging & production deployment	Parshwa, Ishpreet, Neel, Harsh, Nishit, Heer (module completion)	Production deployment complete (P11)
Ishpreet Kaur	202512104	Backend / Database and Query Execution	MongoDB schema, schema discovery, abstraction layer, DB testing/fixes	- (foundational)	Schema & API contract frozen (P4)
Neel Shah	202512114	Backend / Database and Query Execution	FastAPI, query execution engine, API hardening, backend testing/fixes	Ishpreet (schema/DB layer)	Demo-critical pipeline complete (P5)
Harsh Rathod	202512086	Security and Query Validation	Query sanitisation, injection checks, security policy, security test suite	Neel (query execution engine)	Formal testing complete (P8)
Nishit Gal	202512002	Security and Query Validation	Auth/JWT, RBAC, admin panel, auth testing/fixes	Neel (API contract, auth stub)	Parallel track complete (P6)

Member	Student ID	Responsibility Area	Major Tasks	Dependencies	Major Milestone
Parshwa Jain	202512096	Backend / Database and Query Execution	Text-to-query/NLP prompt chain, Hinglish handling, follow-up resolution	Ishpreet (schema discovery contract)	Demo-critical pipeline complete (P5)
Heer Sachdev	202512117	Frontend/UI and UX	Chat interface, result rendering, follow-up UI, UI testing/fixes	Neel (core API endpoints)	Demo-critical pipeline complete (P5)
Bansari Akhani	202512032	Frontend/UI and UX	Chart/table rendering, CSV/Excel export, visualisation testing/fixes	Neel (result payload format)	Parallel track complete (P6)

5. Individual Task Timelines

5.1 Deep Soni (202512089) - Deployment and Post Integration Testing - 202 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	Evaluation approach notes for proposal	4
P2	6-9 Aug	Testing/Deployment module backlog cards created	4
P3	10-16 Aug	Evaluation-set design notes (English/Hinglish pairs)	8
P4	17-27 Aug	AWS environment plan; CI workflow draft (GitHub Actions)	10
P5	28 Aug-11 Oct	150-pair evaluation dataset construction	30
P6	28 Aug-19 Oct	Integration coordination kickoff; test-environment planning	2
P7	20-26 Oct	Lead system integration; merge all modules; staging scaffolding	20
P8	27 Oct-1 Nov	Test plan execution lead; test documentation compiled	33
P9	2-9 Nov	Accuracy measurement run; bug triage coordination; retest sign-off	38
P10	10-13 Nov	Staging deployment lead; AWS config; deployment notes	27
P11	14-15 Nov	Production deployment; final integration sign-off	16
P12	16-18 Nov	Deployment section; overall report compilation lead	10

5.2 Ishpreet Kaur (202512104) - Backend / Database and Query Execution - 219 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	Notes on MongoDB vs relational option for proposal	6
P2	6-9 Aug	Database module backlog cards created	6
P3	10-16 Aug	Collection/document schema requirement notes	12
P4	17-27 Aug	MongoDB schema design; index plan; document map	22
P5	28 Aug-11 Oct	Schema discovery service and metadata cache	60
P6	28 Aug-19 Oct	Database abstraction layer hardening; nested-document relationships	34
P7	20-26 Oct	Database/schema integration merge & verification	8
P8	27 Oct-1 Nov	Database/schema integration tests	20
P9	2-9 Nov	Schema-discovery bug fixes; retest	22
P10	10-13 Nov	Database staging configuration; data seed verification	10
P11	14-15 Nov	Database production verification	6
P12	16-18 Nov	Database module documentation; final report section	13

5.3 Neel Shah (202512114) - Backend / Database and Query Execution - 238 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	API surface sketch for proposal	6
P2	6-9 Aug	Backend module backlog cards created	6
P3	10-16 Aug	Endpoint list & request/response contract draft	12
P4	17-27 Aug	FastAPI service skeleton; auth stub; API contract freeze	20
P5	28 Aug-11 Oct	Query execution engine; timeout & row-limit handling	62
P6	28 Aug-19 Oct	Follow-up/session endpoints; result pagination; API hardening	40
P7	20-26 Oct	Backend/API integration merge & hardening	10
P8	27 Oct-1 Nov	Backend integration tests	22
P9	2-9 Nov	Bug fixes; performance tuning; retest	22
P10	10-13 Nov	API staging configuration & smoke tests	15
P11	14-15 Nov	API production verification	8
P12	16-18 Nov	API documentation (OpenAPI); final report section	15

5.4 Harsh Rathod (202512086) - Security and Query Validation - 212 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	Security risk notes for proposal	4
P2	6-9 Aug	Security module backlog cards created	4
P3	10-16 Aug	Threat model & validation requirement notes	8
P4	17-27 Aug	Query statement whitelist design; sanitisation rules	12
P5	28 Aug-11 Oct	Injection pattern checks; destructive-operation blocking	40
P6	28 Aug-19 Oct	Approval-before-execution UI hook; security policy enforcement	42
P7	20-26 Oct	Security policy integration check across merged build	6
P8	27 Oct-1 Nov	Security test suite build & run	34
P9	2-9 Nov	Security fixes from findings; retest	30
P10	10-13 Nov	Security hardening on staging; secrets review	10
P11	14-15 Nov	Security production verification	6
P12	16-18 Nov	Security documentation; final report section	16

5.5 Nishit Gal (202512002) - Security and Query Validation - 190 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	Role model sketch for proposal	4
P2	6-9 Aug	Auth/Admin module backlog cards created	4
P3	10-16 Aug	Role requirement notes (Admin, Analyst, Viewer)	10
P4	17-27 Aug	Auth data model; JWT flow design; admin wireframe	14
P5	28 Aug-11 Oct	Login/JWT issuance; RBAC middleware	34
P6	28 Aug-19 Oct	Admin panel: DB connections, schema visibility, user mgmt	54
P7	20-26 Oct	Auth/admin integration merge check	6
P8	27 Oct-1 Nov	Auth/RBAC test suite	10
P9	2-9 Nov	Admin panel bug fixes; permission edge cases; retest	30
P10	10-13 Nov	Admin panel/user manual draft	8
P11	14-15 Nov	Auth/admin production verification	4
P12	16-18 Nov	Auth/Admin documentation; final report section	12

5.6 Parshwa Jain (202512096) - Backend / Database and Query Execution - 210 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	Query taxonomy & Hinglish sample collection	6
P2	6-9 Aug	Backlog of NLP tasks drafted; board cards created	6
P3	10-16 Aug	User stories for 6 query categories; NLP requirement notes	14
P4	17-27 Aug	Prompt architecture draft; schema-context format spec	20
P5	28 Aug-11 Oct	Text-to-query prototype (LLM prompt chain) for MongoDB	68
P6	28 Aug-19 Oct	Hinglish query handling; follow-up/context resolution	46
P7	20-26 Oct	NLP module integration merge check	4
P8	27 Oct-1 Nov	NLP unit tests against evaluation query set	8
P9	2-9 Nov	Prompt tuning from evaluation results; retest	20
P10	10-13 Nov	NLP module documentation draft	4
P11	14-15 Nov	NLP production verification	2
P12	16-18 Nov	NLP module documentation; final report section	12

5.7 Heer Sachdev (202512117) - Frontend/UI and UX - 187 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	Chat UI concept notes for proposal	6
P2	6-9 Aug	Frontend module backlog cards created	6
P3	10-16 Aug	Wireframes: chat window, result panel, admin shell	16
P4	17-27 Aug	React component inventory; state management design	16
P5	28 Aug-11 Oct	Chat interface; text/voice input; query preview panel	46
P6	28 Aug-19 Oct	Result table rendering; follow-up query UI; session history	46
P7	20-26 Oct	Frontend integration merge check	4
P8	27 Oct-1 Nov	UI functional tests; cross-browser checks	8
P9	2-9 Nov	UI bug fixes from testing week; retest	18
P10	10-13 Nov	UI polish; screenshots for report	6
P11	14-15 Nov	Frontend production verification	3
P12	16-18 Nov	UI polish; final report section	12

5.8 Bansari Akhani (202512032) - Frontend/UI and UX - 174 h

Phase	Period	Work Package	Hours
P1	27 Jul-5 Aug	Chart/export options notes for proposal	4
P2	6-9 Aug	Visualisation module backlog cards created	4
P3	10-16 Aug	Result-rendering requirement notes (chart types, export formats)	8
P4	17-27 Aug	Chart component selection; export-format spec	10
P5	28 Aug-11 Oct	Basic table + bar chart rendering	30
P6	28 Aug-19 Oct	Line and pie chart rendering; CSV/Excel export implementation	56
P7	20-26 Oct	Visualisation/export integration merge check	2
P8	27 Oct-1 Nov	Visualisation unit tests; UI/testing support	15
P9	2-9 Nov	Export/chart bug fixes; edge cases; retest	20
P10	10-13 Nov	Visualisation documentation draft; UI polish support	10
P11	14-15 Nov	Visualisation production verification	5

Phase	Period	Work Package	Hours
P12	16-18 Nov	Visualisation module documentation; final report section	10

Team total: 1,632 h (202 + 219 + 238 + 212 + 190 + 210 + 187 + 174).

6. Weekly Progress Reports

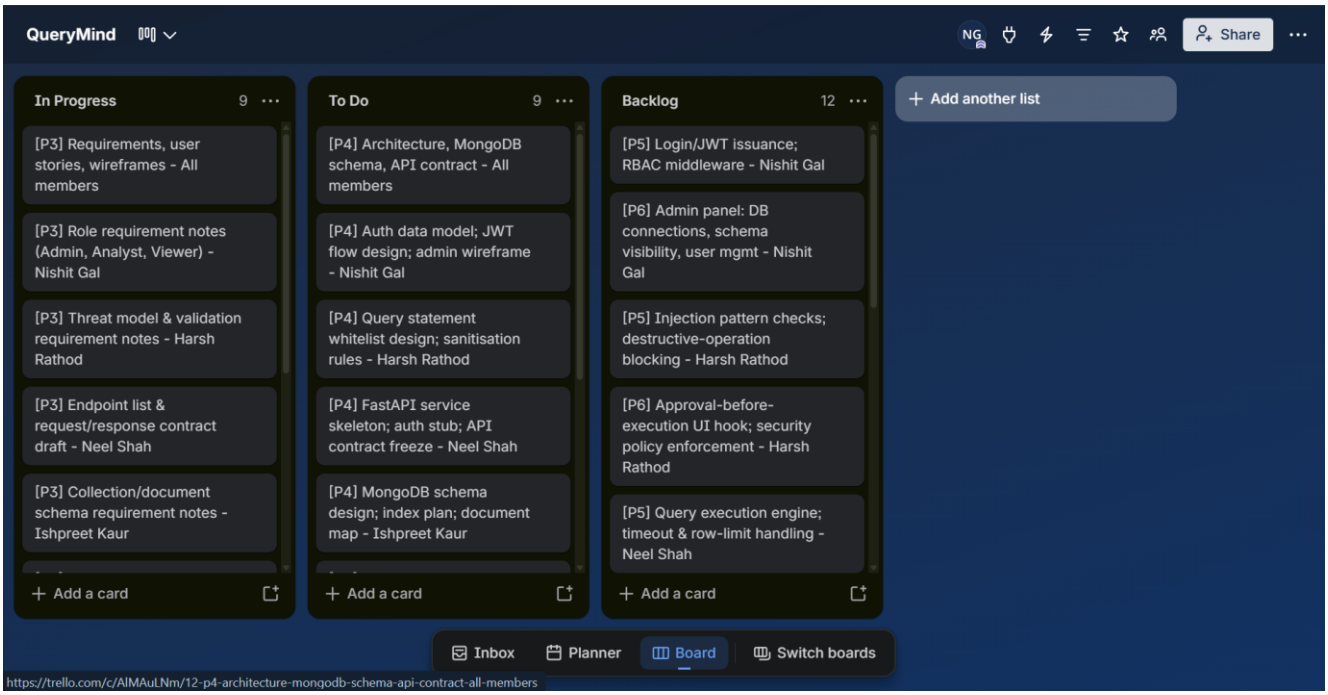
Continuous, one per week, 27 Jul - 15 Nov 2026 (17 weekly cards) - the last progress report covers the week of 9-15 Nov, immediately before the 16-18 Nov final-submission window, so reporting doesn't stop early.

7. Project Management Tools / Task Board

7.1 Board Structure

- Backlog - work packages for phases not yet started.
- To Do - work packages due within the next one to two phases.
- In Progress - work packages inside the current phase.
- Review/Testing - work packages awaiting peer review or QA sign-off before being marked done.
- Done - completed work packages.

7.2 Task Board



7.3 Definition of Done

A code card is done when its pull request is merged into the main branch with tests passing in continuous integration, the API/schema documentation is updated, and the module owner has reviewed it. A document or design card is done when it is reviewed at the weekly team sync and committed to the repository.